FBS Biochemistry — NBME Study Guide

Carbohydrates · Glycosylation · Amino acids & proteins · Protein folding · Enzymes · Vitamins (includes Anki-deck biochemistry)

Red = extra high yield (classic NBME) Black = core content *Italic blue = mnemonic*

1. Carbohydrates

- Classified by carbon number (triose, pentose, hexose) and carbonyl: **aldose** (aldehyde: glucose, galactose, ribose) vs **ketose** (ketone: fructose).
- **D/L = position of –OH on the chiral carbon farthest from the carbonyl** (right = D). Natural sugars are mostly D; natural amino acids are L.
- **Epimers differ at ONE carbon: glucose/galactose (C4), glucose/mannose (C2).** Enantiomers = mirror images. Anomers = differ at the anomeric carbon (α vs β) after ring closure; interconvert by mutarotation.
- **Glycosidic bonds**: O-glycosidic (sugar–OH; disaccharides) vs N-glycosidic (sugar–N; nucleosides, N-linked glycoproteins).
- **Reducing sugar = free anomeric carbon** (Benedict's/Fehling's positive): glucose, fructose, galactose, maltose, lactose. **Sucrose is NON-reducing** (both anomeric carbons used). Trehalose also non-reducing.

Disaccharide	Units & bond	Enzyme
Sucrose	**Glucose α1→β2 fructose (non-reducing)**	**Sucrase**
Lactose	**Galactose β1→4 glucose**	**Lactase (β-galactosidase) — deficiency → bloating, diarrhea, + hydrogen breath test**
Maltose	Glucose α1→4 glucose	Maltase
Trehalose	Glucose α1→1 glucose	Trehalase

- **Polysaccharides**: **glycogen & amylopectin branch with α(1→6)** on α(1→4) chains (glycogen every 8–10 residues — more branched); amylose = unbranched α(1→4); **cellulose = β(1→4), humans can't digest (fiber)**.
- **Liver glycogen maintains blood glucose (has glucose-6-phosphatase); muscle glycogen fuels muscle only.** Liver stores last ~12–24 h of fasting.
- **GAGs**: long unbranched repeating disaccharides (amino sugar + uronic acid), highly **negatively charged** (sulfate, carboxyl) → bind Na^+ and water → gels that cushion/lubricate. Bound to core proteins as **proteoglycans**.

GAG	Key location / fact
Hyaluronic acid	**Only non-sulfated GAG, not attached to core protein; synovial fluid, vitreous, cartilage aggregates**
Heparin	**Only intracellular GAG (mast cell granules); most negatively charged; activates antithrombin**
Chondroitin sulfate	Most abundant; cartilage (aggrecan), bone
Keratan sulfate	Cornea, cartilage
Dermatan sulfate	Skin, vessels, heart valves
Heparan sulfate	Basement membranes, cell surfaces (glomerular charge barrier)

2. Glycosylation (Anki)

	N-linked	O-linked
Residue	**Asparagine (amide N), consensus Asn-X-Ser/Thr (X ≠ Pro)**	**Serine/threonine –OH (hydroxylysine in collagen); no strict consensus**
Where	Begins co-translationally in ER, modified in Golgi	Golgi

How	**En bloc: 14-sugar $Glc_3Man_9GlcNAc_2$ built on dolichol phosphate, transferred by oligosaccharyltransferase (OST)**	One sugar at a time by glycosyltransferases (mucin-type starts with GalNAc)

- **Quality control**: **calnexin/calreticulin** retain misfolded glycoproteins in the ER (glucose trimming/re-adding); persistent misfolding → ERAD → ubiquitin–proteasome.
- **Tunicamycin** blocks dolichol-linked oligosaccharide synthesis → no N-glycosylation.
- **Congenital disorders of glycosylation**: multisystem — hypotonia, developmental delay, abnormal fat pads, inverted nipples, coagulopathy; screen with transferrin isoforms.

3. Amino acids

- All proteins use **L-amino acids**; **glycine is the only achiral amino acid** (R = H; smallest, flexible; every 3rd residue in collagen). D-amino acids in bacterial walls (D-Ala-D-Ala, vancomycin target).
- **Proline**: R group bonds its own amino N (imino acid) → rigid, no backbone H-bond donor → **α-helix breaker**, found at turns; hydroxyproline in collagen. Cis peptide bonds mainly occur before proline.
- **Essential amino acids: PVT TIM HALL** — Phe, Val, Thr, Trp, Ile, Met, His, Arg (conditionally), Leu, Lys.
- **Purely ketogenic: Leucine and Lysine**. Both glucogenic + ketogenic: Phe, Tyr, Trp, Ile, Thr. Ketogenic → acetyl-CoA/acetoacetate; cannot make net glucose (PDH irreversible).
- **Aromatic** (Phe, Tyr, Trp) **absorb UV at 280 nm** (protein assay); nucleic acids at 260 nm; NADH 340 nm.
- **Cysteine** –SH forms **disulfide bonds** (secreted proteins, insulin, antibodies); part of glutathione. Met & Cys contain sulfur.
- **Histidine**: imidazole pKa ≈ 6 (near physiologic pH) → buffering; **proximal His** binds heme Fe^{2+}, **distal His** stabilizes O_2 and sterically limits CO binding (CO affinity ~200–250× O_2). Bohr effect: H^+ protonates His → T state → O_2 unloading.
- Acidic: Asp, Glu (negative at pH 7). Basic: Lys, Arg (positive), His (partly). Polar uncharged: Ser, Thr, Asn, Gln, Cys, Tyr.
- **Sickle cell: Glu → Val at β6 (HbS)**; deoxy-HbS polymerizes (hydrophobic Val). HbC: Glu → Lys (milder, target cells).

Acid–base properties & pI

- **Henderson–Hasselbalch: $pH = pKa + \log([A^-]/[HA])$**. At pH = pKa → 50:50; best buffering within ±1 pH unit of pKa.
- Zwitterion at physiologic pH ($–NH_3^+$, $–COO^-$). **pI = pH with net charge 0**: neutral AA = average of the two pKas (~6); acidic AA = average of the two acidic pKas (low pI); basic AA = average of the two basic pKas (high pI).
- pH < pI → net positive (moves to cathode). pH > pI → net negative (moves to anode). Proteins are least soluble at their pI (precipitate).

4. Proteins: structure, folding & misfolding

Level	Defined by	Bonds
Primary	Amino-acid sequence (N → C)	**Peptide (covalent amide) bonds — planar, rigid, partial double-bond, usually trans**
Secondary	α-helix, β-sheet, turns	**Backbone H-bonds (C=O to N–H)**
Tertiary	3D fold of one chain (domains)	Hydrophobic interactions (main), ionic, H-bonds, disulfides
Quaternary	Multiple subunits (e.g., Hb $\alpha_2\beta_2$)	Same non-covalent ± disulfides

- Insulin: two chains, **3 disulfides** (2 interchain, 1 intrachain on A chain).
- Classification: fibrous (collagen, keratin, elastin — structural, insoluble) vs globular (enzymes, Hb, antibodies — soluble); simple vs conjugated (glyco-, lipo-, metallo-, hemoproteins).
- **Denaturation** (heat, pH, urea, detergents, heavy metals) disrupts 2°–4° structure; primary intact. **Anfinsen**: sequence determines fold (ribonuclease refolds).
- **Chaperones** (Hsp70, Hsp90, chaperonins Hsp60) assist folding; misfolded proteins → ubiquitin → proteasome.

- **Misfolding diseases**: amyloidosis (β-pleated sheet, Congo red → apple-green birefringence), **Alzheimer (Aβ plaques, tau tangles)**, **prion diseases (PrP^{c} α-helix → PrP^{sc} β-sheet; CJD, kuru, mad cow; spongiform, protease-resistant)**, Huntington, Parkinson (α-synuclein), cystic fibrosis ΔF508 (misfolded CFTR degraded), α_1-antitrypsin deficiency (PiZ retained in hepatocyte ER → cirrhosis + emphysema).

5. Enzymes (Anki)

Class	Reaction
1 Oxidoreductases	Electron transfer (dehydrogenases, oxidases)
2 Transferases	Move a group (kinases transfer phosphate, transaminases)
3 Hydrolases	**Break bonds by adding water (proteases, lipases, phosphatases)**
4 Lyases	Break bonds without water/oxidation (decarboxylases, aldolase)
5 Isomerases	Rearrange within a molecule (mutases, epimerases)
6 Ligases	**Join two molecules using ATP (synthetases, carboxylases)**

- **Km = [S] at ½ Vmax (inverse of affinity); Vmax ∝ enzyme amount.** Lineweaver–Burk: x-intercept = −1/Km, y-intercept = 1/Vmax.

Inhibitor	Binds	Km	Vmax	L–B plot
Competitive (reversible)	**Active site (free E)**	**↑**	**—**	**Lines cross on y-axis; overcome by ↑ [S]**
Noncompetitive	**Allosteric site (E and ES)**	**—**	**↓**	**Lines cross on x-axis**
Uncompetitive	ES complex only	↓	↓	Parallel lines
Irreversible	Covalent (aspirin–COX, organophosphates)	—	↓	Like noncompetitive

- **Feedback inhibition**: end product inhibits an early enzyme. **Feedforward activation**: early metabolite activates a later enzyme (F-1,6-BP → pyruvate kinase).

6. Vitamins (water-soluble)

Vitamin	Coenzyme	Function / key enzymes	Deficiency
B1 Thiamine	**TPP**	**Oxidative decarboxylation & transketolase: PDH, α-KG DH, BCKDH, transketolase *(Be APT)***	**Wet beriberi (high-output HF), dry beriberi (neuropathy), Wernicke (confusion, ophthalmoplegia/nystagmus, ataxia), Korsakoff (amnesia, confabulation)**
B2 Riboflavin	**FAD, FMN**	**Redox; succinate DH (Complex II)**	**Cheilosis, magenta tongue, corneal vascularization *(2 C's of B2)***
B3 Niacin	**NAD^{+}, $NADP^{+}$**	**Redox; NADPH for reductive biosynthesis; made from Trp (needs B2, B6)**	**Pellagra: Dermatitis (Casal necklace), Diarrhea, Dementia (± Death)**
B5 Pantothenate	CoA	Acyl transfer; fatty acid synthase	Rare: dermatitis, alopecia, adrenal insufficiency, burning feet
B6 Pyridoxine	**PLP**	**Transamination (ALT, AST), decarboxylation, ALA synthase (heme), glutamate → GABA, glycogen phosphorylase**	**Convulsions, peripheral neuropathy, sideroblastic anemia, cheilosis**
B7 Biotin	**Biotin itself**	**Carboxylation: pyruvate carboxylase, acetyl-CoA carboxylase, propionyl-CoA carboxylase**	**Dermatitis, alopecia, enteritis, lactic acidosis**
B9 Folate	**THF**	**One-carbon transfer: purines, thymidine (dTMP)**	**Megaloblastic anemia, hypersegmented PMNs, ↑ homocysteine, normal MMA, no neuro signs; neural tube defects**

B12 Cobalamin	**Methyl- & adenosyl-cobalamin**	**Methionine synthase; methylmalonyl-CoA mutase**	**Megaloblastic anemia + ↑ homocysteine + ↑ MMA + subacute combined degeneration (dorsal columns, lateral corticospinal, spinocerebellar)**
C Ascorbate	**—**	**Hydroxylation of Pro/Lys (collagen), dopamine β-hydroxylase, $Fe^{3+} \rightarrow Fe^{2+}$ absorption, antioxidant**	**Scurvy: bleeding gums, perifollicular hemorrhage, corkscrew hairs, poor wound healing**

- **Thiamine before glucose** in alcoholics (glucose metabolism consumes TPP → precipitates Wernicke). Dx thiamine: ↑ RBC transketolase activity after adding TPP. Brain & heart (aerobic) suffer first.
- B1 causes: alcoholism (most common US), malnutrition, hyperemesis, bariatric surgery.
- **Pellagra causes beyond diet: Hartnup disease (↓ Trp absorption), carcinoid syndrome (Trp → serotonin), isoniazid (↓ B6).**
- **Niacin drug**: best ↑ HDL, ↓ LDL/TG; side effects flushing (prostaglandin; pretreat with aspirin), hyperglycemia, hyperuricemia (gout).
- **Isoniazid → B6 deficiency → neuropathy; give pyridoxine.** Also penicillamine, OCPs.
- **Biotin: raw egg whites (avidin) or long antibiotics.**
- **Folate deficiency causes: alcoholism, pregnancy, methotrexate, trimethoprim, phenytoin.** Stores last **months**; give before conception (0.4 mg; 4 mg if prior NTD).
- **B12 causes: pernicious anemia (anti-IF/parietal cell antibodies), gastrectomy, terminal ileum disease/resection (Crohn), Diphyllobothrium latum, vegan diet, metformin.** Absorbed with intrinsic factor (parietal cells) in the **terminal ileum**; stores last **years**.
- **Giving folate alone in B12 deficiency corrects the anemia but NOT the neurologic damage.**
- Humans can't make vitamin C (lack L-gulonolactone oxidase). Fat-soluble A, D, E, K are stored (toxicity possible) and need fat absorption.
- **Redox vitamins = B2 + B3; decarboxylation = B1 + B6; carboxylation = biotin; one-carbon = folate (+B12); acyl transfer = B5; hydroxylation = vitamin C.**

7. Rapid-fire NBME pearls

- **Glucose vs galactose = C4 epimers; sucrose non-reducing; glycogen branches α(1→6).**
- **Hyaluronic acid only non-sulfated; heparin only intracellular GAG.**
- **N-linked = Asn-X-Ser/Thr (ER, dolichol, en bloc); O-linked = Ser/Thr (Golgi, one at a time).**
- **Proline breaks helices; glycine achiral; leucine & lysine purely ketogenic; aromatics absorb 280 nm.**
- **Competitive: Km ↑, Vmax same. Noncompetitive: Vmax ↓, Km same.**
- **Wernicke triad + alcoholic → thiamine (before glucose). ↑ MMA → B12, not folate.**
- **Prions = PrP α→β conversion; amyloid stains Congo red (apple-green).**